AI Assistant Coding

Assignment 18.1

Name : K.Navaneeth **HT. No :** 2303A52374 **Batch:** 32

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Code:

```
''' weather feature using open-meteo API and geocoding API to get weather
data based on user input location.
''' import
requests


def get_weather(location):
    location = location.strip()
if len(location) < 3:
        return "Please enter a valid location name with at least 3
characters."

    # Geocoding API to get latitude and longitude from location
geocoding_url = 'https://nominatim.openstreetmap.org/search'
geocoding_params = {         'q': location,
        'format': 'json',
        'limit': 1,
    }
geocoding_headers = {
        # Nominatim requires a valid User-Agent.
        'User-Agent': 'weather-cli/1.0 (student-project)'
    }
```

```
try:
        geocoding_response = requests.get(
geocoding_url,
params=geocoding_params,
headers=geocoding_headers,             timeout=10,
        )
geocoding_response.raise_for_status()
geocoding_data = geocoding_response.json()     except
requests.exceptions.RequestException as exc:
return f"Unable to fetch location data: {exc}"
except ValueError:
        return "Location service returned an invalid response. Please try again
later."
         if not
geocoding_data:
        return "Location not found. Please try again."
         latitude =
geocoding_data[0]['lat']     longitude
= geocoding_data[0]['lon']

    # Open-Meteo API to get weather data     weather_url
= 'https://api.open-meteo.com/v1/forecast'
weather_params = {         'latitude': latitude,
        'longitude': longitude,
        'current': 'temperature_2m,relative_humidity_2m,wind_speed_10m',
    }
try:
        weather_response = requests.get(weather_url,
params=weather_params, timeout=10)
weather_response.raise_for_status()         weather_data =
weather_response.json()     except requests.exceptions.RequestException as
exc:         return f"Unable to fetch weather data: {exc}"     except
ValueError:
        return "Weather service returned an invalid response. Please try
again later."          if 'current' in weather_data:
        current = weather_data['current']
temperature = current.get('temperature_2m')
windspeed = current.get('wind_speed_10m')
humidity = current.get('relative_humidity_2m')
         if temperature is None or windspeed is
None:
```

```
            return "Weather data is incomplete for this location. Please try
again later."
         if humidity is
None:
            return f"The current temperature in {location} is {temperature}°C
with a wind speed of {windspeed} km/h. Humidity data is unavailable."
         return f"The current temperature in {location} is {temperature}°C with
a wind speed of {windspeed} km/h and humidity of {humidity}%."     else:
        return "Weather data not available for this location."
# Example usage if
__name__ == "__main__":
    location = input("Enter a location to get the current weather: ")
print(get_weather(location))
```

Output:

```
(AIAC_env) PS D:\Course\AIAC\labapi253> python weather.py
# Enter a location to get the current weather: warangal
# The current temperature in warangal is 29.1°C with a wind speed of 3.3 km/h
and humidity of 62%.
# (AIAC_env) PS D:\Course\AIAC\labapi253> python weather.py
# Enter a location to get the current weather: SR University
# The current temperature in SR University is 25.9°C with a wind speed of 1.8
km/h and humidity of 79%.
# (AIAC_env) PS D:\Course\AIAC\labapi253> python weather.py
# Enter a location to get the current weather: Devannapet
# The current temperature in Devannapet is 27.7°C with a wind speed of 2.3 km/h
and humidity of 69%.
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Code:

```
''' desigen a currency converter that takes in the amount, the source
currency, and the target currency, and returns the converted amount.
The converter should support at least 5 different currencies and use real-time
exchange rates from an API.
use free api like exchangerate-api.com or openexchangerates.org to get the
exchange rates.
Handle errors such as invalid currency codes, missing API key, and
request failures. ''' import os import requests

API_KEY = os.getenv("4b520c31e5290f6c6d7887d0")
API_URL = "https://api.exchangerate-api.com/v4/latest/"
VALID_CURRENCIES = {"USD", "EUR", "GBP", "JPY", "AUD","INR"}
 def
fetch_exchange_rates(base_currency):
try:
        response = requests.get(f"{API_URL}{base_currency}",
timeout=10)         response.raise_for_status()         data =
response.json()         return data.get("rates", {}), None     except
requests.exceptions.RequestException as exc:
        return None, f"Error fetching exchange rates: {exc}"
 def convert_currency(amount, source_currency,
target_currency):
    source_currency = source_currency.strip().upper()
target_currency = target_currency.strip().upper()
     if source_currency not in
VALID_CURRENCIES:
        return f"Invalid source currency code. Supported currencies: {',
'.join(VALID_CURRENCIES)}"
         if target_currency not in
VALID_CURRENCIES:
        return f"Invalid target currency code. Supported currencies: {',
'.join(VALID_CURRENCIES)}"

    exchange_rates, error = fetch_exchange_rates(source_currency)
if error:
        return error
     if target_currency not in
exchange_rates:
        return f"Exchange rate for {target_currency} not found."

try:
```

```
        converted_amount = amount * exchange_rates[target_currency]
        return f"{amount} {source_currency} is equal to {converted_amount:.2f}
{target_currency}"
except Exception as exc:
        return f"Error during conversion: {exc}"

# Example usage: if
__name__ == "__main__":
    amount = int(input("Enter the amount to convert: "))
source_currency = input("Enter the source currency code (e.g., USD): ")
target_currency = input("Enter the target currency code (e.g., EUR): ")
result = convert_currency(amount, source_currency, target_currency)
print(result)
```

Output:

```
(AIAC_env) PS D:\Course\AIAC\labapi253> python currencyConvert.py
Enter the amount to convert: 100
Enter the source currency code (e.g., USD): USD
Enter the target currency code (e.g., EUR): INR
100 USD is equal to 9405.00 INR
(AIAC_env) PS D:\Course\AIAC\labapi253> python currencyConvert.py
Enter the amount to convert: 100
Enter the source currency code (e.g., USD): INR
Enter the target currency code (e.g., EUR): USD
100 INR is equal to 1.06 USD
(AIAC_env) PS D:\Course\AIAC\labapi253> python currencyConvert.py
Enter the amount to convert: 100
Enter the source currency code (e.g., USD): INR
Enter the target currency code (e.g., EUR): EUR
100 INR is equal to 0.92 EUR
(AIAC_env) PS D:\Course\AIAC\labapi253>
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Code:

```
'''
GitHub Repository Info Fetcher using GitHub REST API.
Fetches repository details including stars, forks, description, and
open issues. ''' import requests


def get_github_repo_info(owner, repo_name):
    """
    Fetch GitHub repository information from the GitHub REST API.

Args:
        owner (str): GitHub repository owner username.
repo_name (str): GitHub repository name.

    Returns:
        dict: Repository information or error message.
    """
    # Input validation     owner = owner.strip() if
owner else ""     repo_name = repo_name.strip() if
repo_name else ""
         if not owner or not
repo_name:
        return "Error: Please provide both owner and repository name."
         if len(owner) < 1 or
len(repo_name) < 1:
        return "Error: Owner and repository name must be at least 1 character
long."

    # GitHub API endpoint
    api_url = f"https://api.github.com/repos/{owner}/{repo_name}"

    headers = {
        'Accept': 'application/vnd.github.v3+json',
        'User-Agent': 'GitHub-Repo-Fetcher/1.0'
    }          try:         response = requests.get(api_url,
headers=headers, timeout=10)
        # Handle rate limit         if
response.status_code == 403:
            remaining = response.headers.get('X-RateLimit-Remaining',
'Unknown')             reset_time = response.headers.get('X-RateLimit-
Reset', 'Unknown')
```

```
            return f"Error: GitHub API rate limit exceeded.
Remaining requests: {remaining}. Reset time: {reset_time}"
        # Handle not found         if response.status_code == 404:
return f"Error: Repository '{owner}/{repo_name}' not found. Please check the
owner and repository name."

        # Handle other HTTP errors         if response.status_code != 200:
return f"Error: Failed to fetch repository data. HTTP Status Code:
{response.status_code}"

        # Parse JSON response
try:
            repo_data     =     response.json()
except ValueError:
            return "Error: GitHub API returned an invalid response. Please try
again later."

        # Extract required fields with safe access
repo_info = {
            'name': repo_data.get('name', 'N/A'),
            'description':  repo_data.get('description')  or  'No  description
available',
            'stars': repo_data.get('stargazers_count', 0),
            'forks': repo_data.get('forks_count', 0),
            'open_issues': repo_data.get('open_issues_count', 0),
            'url': repo_data.get('html_url', 'N/A'),
        }
return repo_info
         except
requests.exceptions.Timeout:
        return "Error: Request timed out. GitHub API is not responding."
except requests.exceptions.ConnectionError:
        return "Error: Network connection error. Please check your
internet connection."     except requests.exceptions.RequestException as
exc:
        return f"Error: Failed to fetch repository data: {exc}"


def display_repo_info(repo_info):
    """
    Display repository information in a formatted manner.
         Args:         repo_info (dict or str): Repository information or
error message.     """
```

```
    if isinstance(repo_info, str):
        # Error message
print(repo_info)     else:
        # Success - display repository details
print("\n" + "="*60)         print("GitHub Repository
Information")         print("="*60)
print(f"Repository Name: {repo_info['name']}")
print(f"URL: {repo_info['url']}")
print(f"Description: {repo_info['description']}")
print(f"Stars: {repo_info['stars']:,}")
print(f"Forks: {repo_info['forks']:,}")
print(f"Open Issues: {repo_info['open_issues']:,}")
print("="*60 + "\n")

# Main execution if
__name__ == "__main__":
    print("GitHub Repository Info Fetcher")
print("-" * 40)
         owner = input("Enter GitHub repository owner (username):
").strip()     repo_name = input("Enter GitHub repository name:
").strip()
         print("\nFetching repository information...")
repo_info = get_github_repo_info(owner, repo_name)
display_repo_info(repo_info)
```

Output:

```
(AIAC_env) PS D:\Course\AIAC\labapi253> python githunInfoFeature.py
GitHub Repository Info Fetcher
----------------------------------------
Enter GitHub repository owner (username): Eshwar1435
Enter GitHub repository name: AIAC-LAB-2072

Fetching repository information...
============================================================
GitHub Repository Information
============================================================
Repository Name: AIAC-LAB-2072
URL: https://github.com/Eshwar1435/AIAC-LAB-2072
Description: No description available
Stars: 0
Forks: 0
Open Issues: 0
============================================================
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Code:

```
'''
Build a News Aggregator  using newsapi.org API to fetch and display news
headlines based on user input topic or category.
Fetch top 5 headlines for a given category (sports, technology, health).
Handle errors like invalid category, missing API key, and request failures.
Display headlines in a user-friendly numbered list format. Include a retry
mechanism if the first request fails.
''' import
os import
time

import requests

API_KEY = os.getenv("NEWS_API_KEY", "6a64706bdc3746f9ba16dda387329610")
BASE_URL = "https://newsapi.org/v2/top-headlines"
VALID_CATEGORIES = {"sports", "technology", "health"}


def fetch_with_retry(params, retries=1, timeout=10):
    last_error = None
     for attempt in range(retries +
1):         try:
            response = requests.get(BASE_URL, params=params,
timeout=timeout)             response.raise_for_status()             return
response.json(), None         except requests.exceptions.RequestException as
exc:
            last_error = exc
if attempt < retries:
```

```
                time.sleep(1)
     return None, f"Unable to fetch news headlines after {retries +
1} attempt(s): {last_error}"
 def
get_news_headlines(category):
    category = category.strip().lower()
     if category not in
VALID_CATEGORIES:
        return "Invalid category. Please choose from: sports, technology,
health."
     if not API_KEY or API_KEY ==
"YOUR_API_KEY_HERE":
        return "Missing API key. Set NEWS_API_KEY environment variable or
update API_KEY in the script."
     params =
{
        "category": category,
        "pageSize": 5,
        "country": "us",
        "apiKey": API_KEY
    }      data, request_error = fetch_with_retry(params, retries=1,
timeout=10)     if request_error:
        return request_error
     if not isinstance(data,
dict):
        return "News service returned an invalid response. Please try
again later."      if data.get("status") != "ok":
        return f"Error fetching news headlines: {data.get('message', 'Unknown
error')}"
     headlines = data.get("articles",
[])     if not headlines:
        return f"No headlines found for category: {category}"
     result_lines = [f"Top 5 Headlines for
{category.capitalize()}:"]     for i, article in
enumerate(headlines, start=1):
        title = article.get("title") or "Untitled headline"
result_lines.append(f"{i}. {title}")
     return
"\n".join(result_lines)
 if __name__ ==
"__main__":
    category = input("Enter a category (sports, technology, health): ")
```

```
    print(get_news_headlines(category))
```

Output:

```
(AIAC_env) PS D:\Course\AIAC\labapi253> python newsHeadline.py
Enter a category (sports, technology, health): sports
Top 5 Headlines for Sports:
1.    MLB 2026 season preview: Rankings, every team's playoff odds - ESPN
2.    Syracuse finalizes deal with Gerry McNamara to become next head coach:
‘I’m ready for it’ - Syracuse.com
3.    UK booster asks President Capilouto to reverse Mitch Barnhart’s $950K
role - On3
4.    Post-Free Agency NFL Power Rankings: Bengals, Commanders Are Contenders
Again - Sports Illustrated
5.    'If he'd known what condition the bike was in, he wouldn't have been
descending like that' – Tadej Pogačar won Milan-San Remo on a cracked frame
with disc brake rub - Cyclingnews
(AIAC_env) PS D:\Course\AIAC\labapi253> python newsHeadline.py
Enter a category (sports, technology, health): technology
Top 5 Headlines for Technology:
1.    Just get a new Samsung phone? Turn off these 6 features immediately -
Android Authority
2.    We Didn't Expect This Minecraft Spin-Off to Get a PS5 Sequel - Push
Square
3.    Nintendo Cuts Switch 2 Output by Over 30% on Weak US Sales -
Bloomberg.com 4. Android Auto is breaking for Pixel and Samsung users, and no
one knows why - Android Authority
5. Claude can now automate your entire desktop, but with a serious limitation -
XDA
(AIAC_env) PS D:\Course\AIAC\labapi253>
```